

Canted antiferromagnetism in a van der Waals metallic material TaNi₂Te₂

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Since the discovery of intrinsic ferromagnetism in CrI₃ and Cr₂Ge₂Te₆, research has been initiated to explore diverse new van der Waals (vdW) magnetic materials. In this study, we successfully synthesized high-quality single crystals of the vdW material TaNi₂Te₂ using the chemical vapor transport method. Subsequently, we mechanically exfoliated the crystals into thin layers and investigated their physical properties. Our results reveal the presence of a rare canted antiferromagnetic behavior in TaNi₂Te₂, which is accompanied by remarkable metallic characteristics and elevated carrier mobility of 1.05 cm²/Vs. This study offers a approach to achieving canted antiferromagnetism in two-dimensional vdW systems, and provides intriguing possibilities in spintronics and electronic devices.

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I. INTRODUCTION

As the extensively studied 2D material, graphene has garnered widespread acclaim for its exceptional properties [1], including high carrier mobility [2], mechanical strength [3], and thermal conductivity [4]. In the past two decades, the field of materials science has experienced a burgeoning interest in two-dimensional (2D) materials, largely due to the successful exfoliation and application of graphene [5]. The distinctive layered structure of these materials has revealed electrical [6], optical [7], and magnetic properties [8,9]. Furthermore, their facile exfoliation has facilitated convenient processing into diverse devices using cutting-edge micro and nanoscale fabrication techniques [10–12]. Since the groundbreaking revelation in 2017 of long-range magnetic order persisting within the confines of an atomic layer thickness in CrI₃ [9] and Cr₂Ge₂Te₆ [8], an increasing number of two-dimensional magnetic materials have been reported. Examples include MPX₃ ($M = \text{Mn, Fe, Co, Ni}$; $X = \text{S, Se}$) [13–15], CrX₃ ($X = \text{F, Cl, Br, I}$) [9,16–18], and Fe₃GeTe₂ [19]. These investigations have emerged as an innovative avenue for delving into the potential applications of quantum magnetism and spintronics.

While significant progress has been made in understanding conventional ferromagnetic and antiferromagnetic materials, the exploration of canted antiferromagnetic materials in van der Waals (vdW) systems remains relatively limited. The unique characteristics of canted antiferromagnetic materials, such as tunable magnetic properties and efficient spin transport, make them promising candidates for future spintronic applications [20,21]. However, the lack of reported two-dimensional canted antiferromagnetic materials limits our ability to fully harness their potential in practical devices. In this paper, we report a two-dimensional material TaNi₂Te₂ with canted antiferromagnetism. Generally, the spin canting

of the material can lead to the cant of the magnetic moments that were originally arranged in the antiferromagnetic order in the lattice, resulting in canted antiferromagnetism [22]. This spin canting is mainly caused by superexchange interaction and Dzyaloshinskii-Moriya (DM) antisymmetric exchange [23–27]. TaNi₂Te₂ serves as an exemplary 2D antiferromagnetic material, where the Ta atom resides within Ni and Te octahedra, forming one-dimensional (1D) chains. These chains are further interconnected through Te atoms, resulting in a layered van der Waals structure [28–30]. In this investigation, we successfully synthesized high-quality TaNi₂Te₂ crystals using a two-step chemical vapor transport (CVT) method. The crystal structure and chemical composition were thoroughly characterized using techniques such as x-ray powder diffraction (XRD), scanning electron microscopy (SEM), energy dispersive x-ray spectroscopy (EDS), selected area electron diffraction (SAED), and high-resolution scanning transmission electron microscopy (HRSTEM). Additionally, we conducted electrical and magnetic measurements to ascertain the fundamental physical properties of TaNi₂Te₂ and explore its canted antiferromagnetism. This presents prospects for the advancement of two-dimensional electronic devices.

II. EXPERIMENTAL METHODS

A. Synthesis and characterization of crystal TaNi₂Te₂

Single crystals of TaNi₂Te₂ were synthesized in our research using a two-step process. In the first step, polycrystalline powder samples of TaNi₂Te₂ were prepared through high-temperature solid-state reactions. The mixture of Ta, Ni, and Te elements, with a molar ratio of 1:2:2, was transferred to a silica tube and sealed under vacuum. Subsequently, the sealed tube was heated to 900 °C for three days in a Muffle furnace. The second step involved chemical vapor transport. An appropriate amount of iodine was added to the resulting powder to facilitate vapor transport. The

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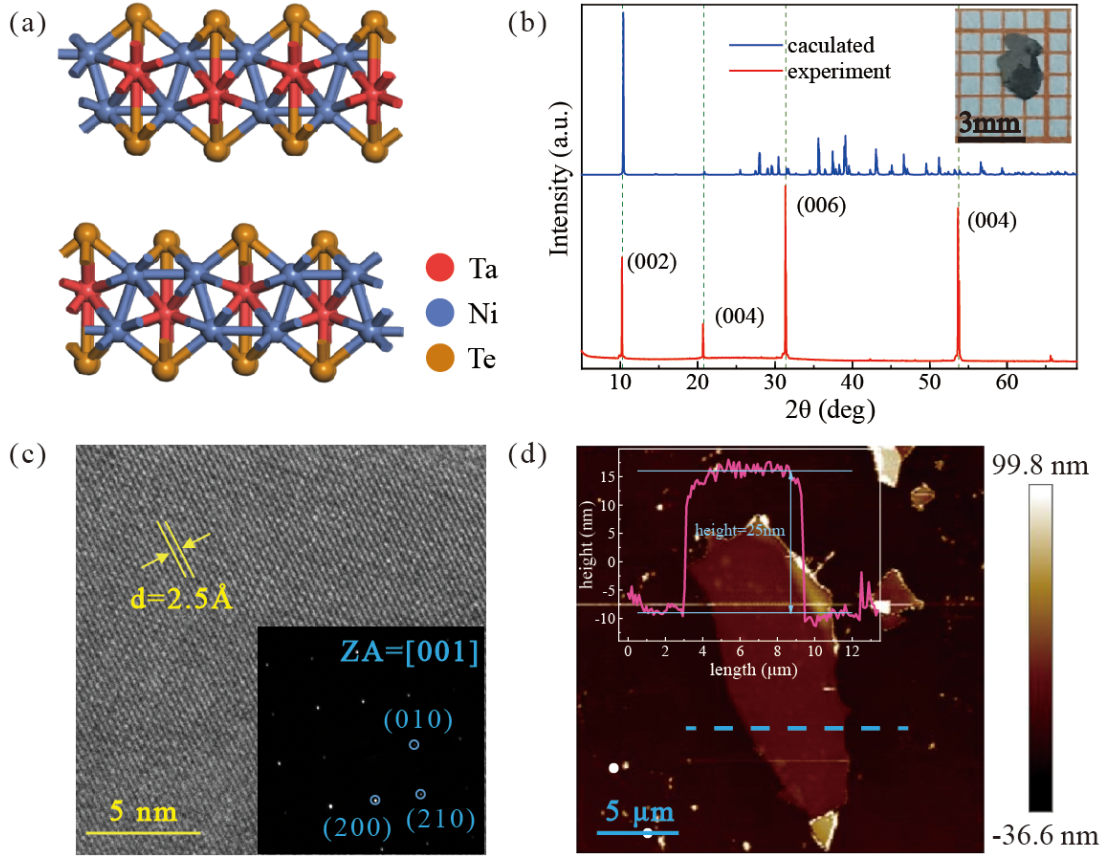


FIG. 1. Crystal structure of TaNi_2Te_2 . (a) Calculated models of TaNi_2Te_2 . (b) X-ray diffraction patterns of TaNi_2Te_2 , inset shows the photo of the single crystal TaNi_2Te_2 . (c) High-resolution scanning transmission electron microscopy image of single crystal TaNi_2Te_2 , inset shows selected area electron diffraction pattern along the $[001]$ axis. (d) Atomic force microscopy image of TaNi_2Te_2 thin flake, inset shows height gradient at the dotted line.

mixture was then sealed into a new silica tube under vacuum and heated in a two-zone tube furnace with a temperature gradient of 950–850 °C for two weeks. As a result, single crystals of TaNi_2Te_2 were obtained at the cold end of the tube.

The crystal structure was characterized by using powder x-ray diffraction (XRD, Ultima IV) with a $\text{Cu K}\alpha$ radiation source. The thickness of the exfoliated samples on SiO_2/Si was characterized by using atomic force microscopy (AFM, Dimension Icon). The chemical compositions were characterized by using scanning electron microscopy (SEM, FEL Apreo) and energy-dispersive x-ray spectroscopy (EDS, X Flash 6130). The chemical stoichiometry and valence states were characterized by using x-ray photoelectron spectroscopy (XPS, Kratos AXIS SUPRA+).

B. Fabrication of Hall devices

We fabricated a Hall bar on a silicon substrate with a 285 nm thick SiO_2 layer using ultraviolet lithography. The conducting media consisted of an 8 nm thick layer of Ti followed by a 25 nm thick layer of Au, which were deposited onto the Hall electrode through magnetron sputtering. The TaNi_2Te_2 sample was synthesized and then mechanically exfoliated into thin films. Subsequently, these thin films were transferred onto the fabricated device by a dry-transfer technique with an accurate microscopic transfer platform (Metatest, E1-T).

C. Electric and magnetic properties

The electronic transport measurements were performed in a physical property measurement system (PPMS) fabricated by Quantum Design, in which the basic temperature and magnetic field of the platform can reach 1.8 K and 9 T, respectively. The testing current was 100 μA , and the magnetic field was applied perpendicular to the sample.

The temperature-dependent magnetization (MT) and field-dependent magnetization (MH) of the bulk crystal were characterized using a superconducting quantum interference device (SQUID), specifically the Quantum Design MPMS.

III. RESULT AND DISCUSSION

A. Crystal structure

TaNi_2Te_2 possesses a layered structure with space group Pnma (62), as depicted in Fig. 1(a). The layers are stacked along the c axis through weak van der Waals bonds involving Te atoms. Each individual layer of TaNi_2Te_2 comprises five atomic layers, with Te atoms forming the outermost layer, Ni atoms occupying the second and fourth layers, and Ta atoms sandwiched in the middle, forming quasi-one-dimensional atomic chains. To characterize the structure and crystallinity of the synthesized TaNi_2Te_2 sample, we employed x-ray powder diffraction (XRD), and compared the result with the

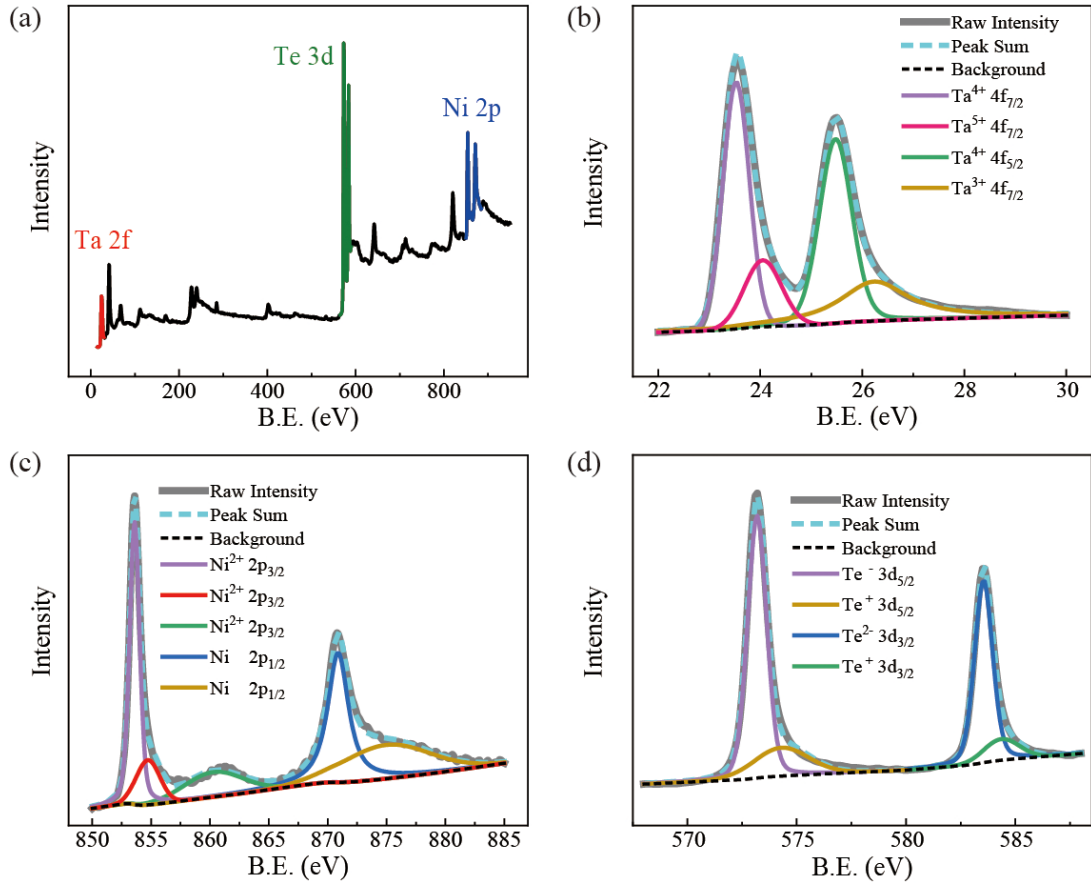


FIG. 2. X-ray photoelectron spectroscopy spectra of (1) TaNi_2Te_2 , (b) Ta, (c) Ni, and (d) Te.

theoretical calculation as shown in Fig. 1(b). The positions of the four principal peaks exhibit a high degree of conformity with the corresponding calculated values, which are indexed to the (002), (004), (006), and (0010) crystal planes, respectively. Notably, these peaks display a marginal leftward deviation, falling well within an acceptable range of error. The presence of sharp and strong XRD peaks indicates the high quality of the synthesized single crystal TaNi_2Te_2 . The inset of Fig. 1(b) shows the photo of the single crystal TaNi_2Te_2 . The macroscopic dimensions are approximately 2×3 mm, and its surface exhibits a metallic luster. In addition to XRD, we utilized scanning electron microscopy (SEM) and energy dispersive x-ray spectroscopy (EDS) to analyze the chemical composition of our sample. The EDS spectra (see Fig. S1 in the Supplemental Material [49]) shows that the stoichiometric ratio of Ta:Ni:Te inferred from this point is approximately 1:2.06:2.12, which agrees well with the expected ratio for TaNi_2Te_2 . The HRSTEM patterns shown in Fig. 1(c) also reveal that the TaNi_2Te_2 sample possesses high crystallinity with $d = 2.5$ Å. The SAED pattern shown in the inset of Fig. 1(c) coincides well with the simulated pattern from the reported crystal structure. We used tape to mechanically exfoliated crystalline TaNi_2Te_2 and subsequently transferred it onto a pristine silicon wafer. Following the successful transfer, the morphology of the obtained material is analyzed using atomic force microscopy (AFM). This yield topography displayed in Fig. 1(c) exhibits a smooth and clean surface, and the AFM height profile [Fig. 1(d)] characterizes the flake

thickness to be ~ 25 nm. This proves that TaNi_2Te_2 has a vdW two-dimensional layered structure. In addition, we conducted Raman spectroscopy measurements on samples synthesized about half a year ago and exposed to air, as well as on recently synthesized samples (see Fig. S2 in the Supplemental Material [49]). These two samples share the same peak position, with only some differences in intensity, which proves that TaNi_2Te_2 is not air sensitive.

We characterized the valence state for each composition in the ternary compound TaNi_2Te_2 by x-ray photoelectron spectroscopy (XPS). Clear characteristic signals subject to the three constituent elements can be readily identified in the total spectrum presented in Fig. 2(a). Figures 2(b)–2(d) show the zoomed-in plots near the peak positions of the three elements. In Fig. 2(b), two dominant peaks of Ta exist at 23.55 eV and 25.5 eV, consistent with $4f_{7/2}$ of Ta^{4+} and $4f_{5/2}$ of Ta^{4+} , respectively [31,32]. There are two satellite peaks around these main peaks consistent with $4f_{7/2}$ of Ta^{5+} and $4f_{7/2}$ of Ta^{3+} [33,34]. In Fig. 2(c), the main peak at 853.6 eV is related to $2p_{3/2}$ of Ni^{2+} , two satellite peaks at 854.7 and 860.7 eV are resolved as the signal of $2p_{3/2}$ of Ni^{2+} , while other secondary peaks located at 870.9 and 875.5 eV are both indexed to $2p_{1/2}$ of Ni [35–37]. In Fig. 2(d), there are two dominant peaks of Te at 573.2 and 583.6 eV, which correspond to $3d_{5/2}$ of Te^- and $3d_{3/2}$ of Te^{2-} , respectively. Their two satellite peaks represent $3d_{5/2}$ and $3d_{3/2}$ of Te^+ , which indicate the presence of a minor degree of surface oxidation in the sample [38,39].

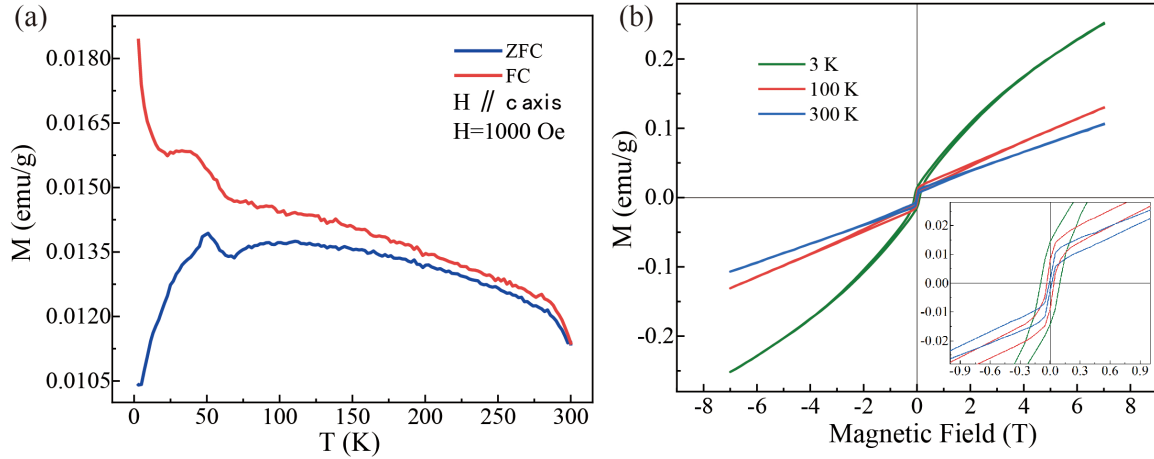


FIG. 3. Magnetic properties of TaNi_2Te_2 . (a) Temperature dependence of the magnetization (M) under ZFC and FC conditions with the field of 1000 Oe applied along the c axis. (b) Magnetic field dependence of the magnetization (M) at different temperature with the field applied along the c axis, inset shows the details under the magnetic field within ± 1 T.

B. Magnetic properties

We conducted magnetic measurements on the TaNi_2Te_2 crystal using the SQUID system to study its magnetic properties. The temperature dependence of magnetization under zero field cooling (ZFC) and field cooling (FC) conditions, with an external magnetic field of 1000 Oe applied along the c axis, is illustrated in Fig. 3(a). The ZFC curve reaches a maximum at 50 K, which shows an antiferromagnetic (AFM) transition at $T_N = 50$ K. Below this T_N , ZFC and FC curves show different trends at about 35 K; this can be attributed to the intrinsic pinning effect [40]. We reduced the magnetic field to 200 Oe and repeated the M-T measurement under FC conditions [see Fig. S3(c) in the Supplemental Material [49]] [41]. The rise of the curve at low temperatures has decreased significantly. In the FC curve, there is a wide platform on the left side of T_N , which may be due to spin-spin correlation typical of low-dimensional magnetic systems [42]. In paramagnetic range at high temperature, the FC curve is slightly higher than the ZFC curve, while they have the same tendency. There is a slight turning point at 70 K, which might be caused by mixed ferromagnetic phase [43]. Figure 3(b) shows the magnetic field dependence of magnetization (M) at different temperatures of 3, 100, and 300 K. The enlarged view [inset of Fig. 3(b)] at lower field ranges shows minor hysteresis from 3 K to 300 K. The coercivity (H_C) decreases with temperature increasing. Here H_C in 3, 100, and 300 K is 966, 368, and 148 Oe, respectively. When temperature is increasing, the atomic vibration caused by thermal effect would affect the arrangement of magnetic moment, making TaNi_2Te_2 more susceptible to magnetization, resulting in the reduction of H_C [44,45]. Under high magnetic field, the magnetization increases slowly and linearly with the increasing field, and its remnant magnetization (M_r) also decreases with temperature increasing. This kind of weak isothermal magnetization and strong field dependence shows a canted antiferromagnetic state [46–48]. When the external magnetic field is parallel to the ab plane, The M-T curve also exhibits an antiferromagnetic transition at 50 K, while the M-H curve does not reflect significant magnetic behavior [see Figs. S3(a) and S3(b) in the Supplemental

Material [49]], which prove that the antiferromagnetic order of TaNi_2Te_2 is out of plane.

C. Electronic transport properties

We conducted electrical measurements on TaNi_2Te_2 to further investigate its electrical properties. The temperature-dependent electrical resistivity (ρ_{xx}) of TaNi_2Te_2 was measured using a four-terminal method, as illustrated in the inset of Fig. 4(a). The exfoliated thin film crystal of TaNi_2Te_2 was transferred onto the Hall electrode, and a current was applied to the middle pair of electrodes, which had a width of 8 μm . In Fig. 4(a), it can be observed that the resistance remains relatively constant at temperatures below 25 K but increases as the temperature rises above this threshold till room temperature. Moreover, the resistivity of the sample is on the order of $10^{-7} \Omega \text{cm}$, indicating a metallic behavior in TaNi_2Te_2 . The residual resistivity ratio [$\text{RRR} = \rho_{xx}(300 \text{ K})/\rho_{xx}(3 \text{ K})$] is ≈ 97 , which indicates a decent crystal quality. Then, we performed magnetotransport measurements at different temperatures using the device shown in the inset of Fig. 4(a) by applying an out-of-plane magnetic field perpendicular to the ab plane and along the c axis. The maximum magnetoresistance (MR) calculated by formula (1) is 63.6% in 3 K as shown in Fig. 4(b):

$$MR = \frac{\rho_{xx}(H_{\max}) - \rho_{xx}(H_0)}{\rho_{xx}(H_0)} \times 100\%. \quad (1)$$

With the temperature increasing, the magnetoresistance decreases rapidly, with MR at 30 K and 100 K being 21.2% and 0.5% (see Fig. S4 in the Supplemental Material [49]), respectively. At high temperatures, MR is almost negligible. To further investigate the nature of the charge carriers, we performed Hall effect measurements in various temperatures. Figure 4(c) shows a linear dependence of the Hall resistivity (ρ_{xy}) on the applied magnetic field in 3 K, and maintains similar linearity at 30 K and 100 K (see Fig. S4 in the Supplemental Material [49]). We calculated the directional

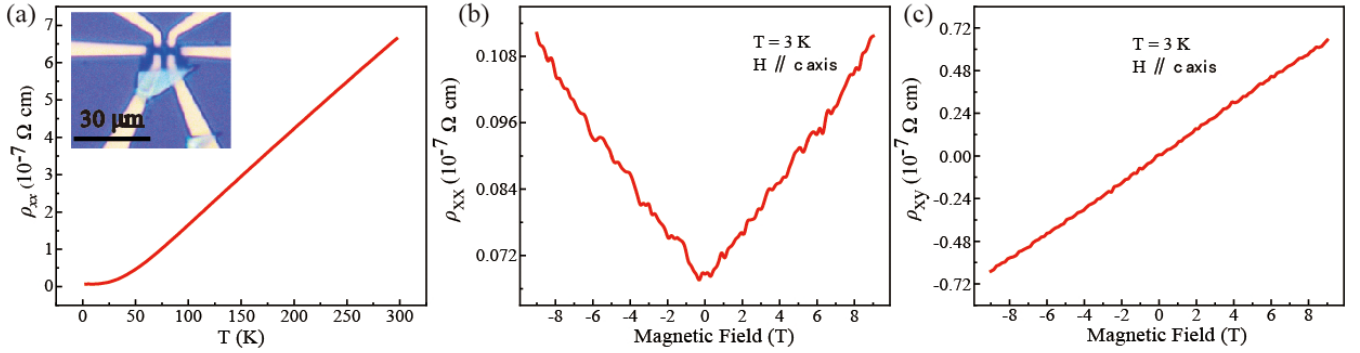


FIG. 4. Electronic transport properties of TaNi_2Te_2 . (a) Temperature dependence of the resistivity, inset shows the optical image of the electronic device. (b) Magnetic field dependence of the resistivity (ρ_{xx}) at low temperature (3 K) with the magnetic field applied along the c axis. (c) Magnetic field dependence of the Hall resistivity (ρ_{xy}) at low temperature (3 K) with the magnetic field applied along the c axis.

carrier mobility (μ) of TaNi_2Te_2 by combining formulas (2) and (3):

$$\sigma_{xx} = \frac{\rho_{xx}(H_0)}{[\rho_{xx}(H_0)]^2 + [\rho_{xy}(H_0)]^2}, \quad (2)$$

$$\mu = \sigma_{xx} \frac{d\rho_{xy}}{dH}. \quad (3)$$

At the temperature of 3 K, the longitudinal conductivity (σ_{xx}) is calculated as $1.45 \times 10^{10} \Omega^{-1}\text{m}^{-1}$. Here the derivative $d\rho_{xy}/dH$ is determined to be $7.23 \times 10^{-11} \Omega\text{m T}^{-1}$. To sum up, the obtained carrier mobility $\mu = 10484.5 \text{ cm}^2/\text{Vs}$. We compared this mobility with some conventional 2D magnetic materials, and it is higher than the vast majority of magnetic metallic materials except of CdTe_3 (see Table S1 in the Supplemental Material [49]), demonstrating that TaNi_2Te_2 is a high mobility magnetic van der Waals system.

IV. CONCLUSION

In this study, we have successfully synthesized a 2D van der Waals material named TaNi_2Te_2 and thoroughly characterized its chemical composition and crystal structure. By means of our investigations into its electrical and magnetic properties, we have discovered that it exhibits relatively uncommon canted antiferromagnetism, and serves as a metallic material with elevated carrier mobility. In light of these outcomes, TaNi_2Te_2 displays great potential as a captivating contender for the advancement of innovative two-dimensional electronic devices.

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